

# What is a Lattice?

Lattices in cryptography are defined as sets of all integer linear combinations of certain  $n$ -dimensional real space basis vectors. Even more simply, a lattice contains every combination that can be formed by adding together integer multiples of the basis vectors. The dimension ( $n$ ) of the lattice depends on the amount of basis vectors, which consist of  $n$  amount of real numbers.

As the number of basis vectors increases, the lattice gets exponentially more complex. This matters because solving certain computational problems on high-dimensional lattices are believed to be extremely difficult. These problems form the security foundation of modern lattice-based cryptography.

● b1

1

↕

0

↕

● b2

0

↕

1

↕

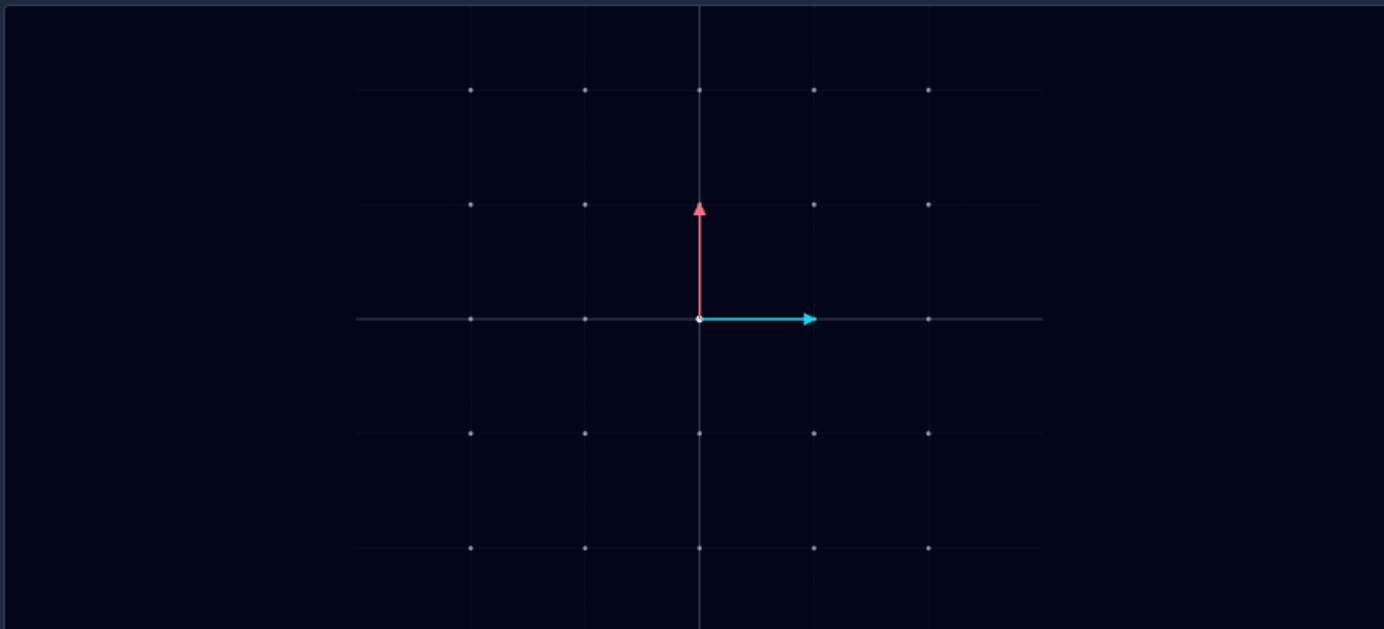
+ Add basis vector

Show ± 

2

▼

 combinations



The dots represent the integer combination of the basis vectors- in other words, they represent the lattice. The colored arrows are the basis vectors. Hover over point to highlight it, click to see what it's made of.